ILAE 2017 Classification of Seizure Types and Definitions

1. Overview and Operational Definition

In 2017, the International League Against Epilepsy (ILAE) presented an updated operational classification of seizure types. Epilepsy is clinically defined by any of the following conditions: (1) at least two unprovoked or reflex seizures occurring more than 24 hours apart; (2) one unprovoked seizure and a probability of further seizures similar to the general recurrence risk after two unprovoked seizures (at least 60%) occurring over the next 10 years; or (3) diagnosis of an epilepsy syndrome.

The classification divides seizures by type of onset: focal onset, generalized onset, and unknown onset. A seizure is classified as having focal onset when the initial activation of neuronal networks is limited to one cerebral hemisphere.

2. Focal Seizure Subclassification

Focal seizures are categorized based on awareness during the event. In Focal Aware Seizures (FAS, previously termed simple partial seizures), consciousness and responsiveness remain fully intact. In Focal Impaired Awareness Seizures (FIAS, previously complex partial seizures), awareness is impaired at any point during the seizure.

Focal seizures are further subclassified by motor or non-motor onset signs. Motor manifestations include automatisms (involuntary repetitive semi-purposeful motor behaviors like lip-smacking or hand-fumbling), atonic, clonic, epileptic spasms, hyperkinetic movements (thrashing or pedaling commonly originating from the frontal lobe), and tonic posturing.

Non-motor focal onset signs include autonomic symptoms (epigastric rising sensation, flushing, pupillary changes), cognitive phenomena (deja vu, jamai vu, aphasia, micro/macropsia), emotional changes (sudden paroxysmal fear or anxiety, ictal laughter/gelastic), and sensory auras (somatosensory tingling, visual hallucinations, olfactory or gustatory illusions).

3. Progression and Generalized Seizure Types

A focal seizure may propagate across the corpus callosum and bilateral networks, termed a Focal to Bilateral Tonic-Clonic Seizure (FBTCS, replacing secondary generalized seizure).

Generalized seizures originate at some point within and rapidly engage bilaterally distributed cerebral networks. Generalized motor seizures include tonic-clonic, clonic, tonic, atonic, myoclonic, and myoclonic-atonic variants.

Generalized non-motor seizures are typical absence seizures (characterized by sudden behavioral arrest and 3 Hz generalized spike-and-wave on scalp EEG), atypical absence (often with slower, less abrupt onset/offset and <2.5 Hz spike-wave activity), and myoclonic absence.

4. Scalp Electroencephalography (EEG) Signatures

Scalp EEG provides vital neurophysiological confirmation. Focal epileptiform discharges manifest as localized sharp waves (70-200 ms duration) or spikes (20-70 ms duration), frequently accompanied by localized aftercoming slow waves.

Anterior temporal spikes (maximal at F7/T3 or F8/T4 electrodes) strongly correlate with hippocampal sclerosis and mesial temporal lobe epilepsy. Frontal spikes and fast paroxysmal rhythms correlate with focal cortical dysplasia in frontal lobe epilepsy.

Childhood absence epilepsy displays synchronous, symmetrical 3 Hz generalized spike-wave complexes on normal background activity. Hypsarrhythmia, characterized by chaotic, high-voltage, disorganized activity with multifocal sharp waves, is the pathognomonic EEG hallmark of West syndrome (infantile spasms).